Preview: y2003_q17 (v1)

Alternating rows: starts and ends the same, so one more of that kind; then split a group

Part A: Each question is worth 3 points

1. A bar code is made of 9 black and white bars, black and white taking turns. The first and the last bar are black. Each black bar is either wide or narrow. There are 2 more white bars than wide black bars.

How many narrow black bars are there?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

2. A bar code is made of 13 black and white bars, black and white taking turns. The first and the last bar are black. Each black bar is either wide or narrow. There are 4 more white bars than wide black bars.

How many narrow black bars are there?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

3. Nina builds a straight fence from 21 pieces: posts and boards take turns, and the fence starts and ends with a post.
How many boards are there?

A. 21 ***B.*** 12 ***C.*** 11 ***D.*** 10 ***E.*** 8

4. Mia builds a straight fence from 21 pieces: posts and boards take turns, and the fence starts and ends with a post.
How many boards are there?

A. 21 ***B.*** 12 ***C.*** 11 ***D.*** 10 ***E.*** 8

Part B: Each question is worth 4 points

5. A bar code is made of 11 black and white bars taking turns, with a black bar at each end. Each black bar is wide or narrow. There are 4 more white bars than wide black bars.

How many wide black bars are there?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 5 ***E.*** 9

6. A bar code is made of 31 black and white bars taking turns, with a black bar at each end. Each black bar is wide or narrow. There are 10 more white bars than wide black bars.

How many wide black bars are there?

A. 5 ***B.*** 4 ***C.*** 3 ***D.*** 2 ***E.*** 1

7. Nina threads 19 beads on a string: red and blue take turns, and the string starts and ends with a red bead. Each red bead is big or small. There are 3 more blue beads than big red beads.
How many big red beads are there?

A. 6 ***B.*** 7 ***C.*** 8 ***D.*** 9 ***E.*** 11

8. Tom threads 23 beads on a string: red and blue take turns, and the string starts and ends with a red bead. Each red bead is big or small. There are 8 more blue beads than big red beads.
How many big red beads are there?

A. 3 ***B.*** 4 ***C.*** 5 ***D.*** 7 ***E.*** 9

Part C: Each question is worth 5 points

9. A bar code is made of 16 black and white bars taking turns. It starts with a black bar and ends with a **white** bar. Each black bar is wide or narrow. There are 2 more white bars than wide black bars.
How many narrow black bars are there?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 4 ***E.*** 5

10. A bar code is made of 14 black and white bars taking turns. It starts with a black bar and ends with a **white** bar. Each black bar is wide or narrow. There are 4 more white bars than wide black bars.
How many narrow black bars are there?

A. 4 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

Answer Key

1. **D**	2. **E**	3. **D**	4. **D**	5. **B**	6. **A**
7. **A**	8. **A**	9. **C**	10. **A**		

Solutions

1. D (3) — Black is at both ends, so there are 5 black and 4 white bars. Wide: 4 – 2 = 2. Narrow: 5 – 2 = 3. [2003 Q17 · narrow_bars]

2. E (5) — Black is at both ends, so there are 7 black and 6 white bars. Wide: 6 – 4 = 2. Narrow: 7 – 2 = 5. [2003 Q17 · narrow_bars]

3. D (10) — Post, board, post, ..., post: there is one more post than boards. 21 = 11 + 10, so 11 posts and 10 boards. [2003 Q17 · fence]

4. D (10) — Post, board, post, ..., post: there is one more post than boards. 21 = 11 + 10, so 11 posts and 10 boards. [2003 Q17 · fence]

5. B (1) — There are 6 black and 5 white bars. Wide: 5 – 4 = 1. [2003 Q17 · wide_bars]

6. A (5) — There are 16 black and 15 white bars. Wide: 15 – 10 = 5. [2003 Q17 · wide_bars]

7. A (6) — There are 10 red and 9 blue beads. Big red: 9 – 3 = 6. [2003 Q17 · necklace]

8. A (3) — There are 12 red and 11 blue beads. Big red: 11 – 8 = 3. [2003 Q17 · necklace]

9. C (2) — Starting black and ending white, the bars come in pairs: 8 black and 8 white. Wide: 8 – 2 = 6. Narrow: 8 – 6 = 2. [2003 Q17 · ends_differ]

10. A (4) — Starting black and ending white, the bars come in pairs: 7 black and 7 white. Wide: 7 – 4 = 3. Narrow: 7 – 3 = 4. [2003 Q17 · ends_differ]